Review Questions 1-20

1. b. The investigator doesn't have to get a warrant.

2. True

3. True

4. a. You begin to take orders from a police detective without a warrant or subpoena.

5. False

6. a. Coordinate with the HAZMAT team.

7. Three rules for a forensic hash: It must be consistent (same input = same output). It must be unique (different input = different output). It must be irreversible (cannot reconstruct original data from the hash).

8. A collision occurs when two different files produce the same hash value.

9. Three items in an initial-response field kit: Write blockers, Evidence bags and labels, Camera or video recorder

10. To avoid contaminating, altering, or damaging evidence. Digital evidence can be easily changed.

11. True

12. b. Disconnecting power.

13. You should videotape or sketch: The entire room layout, Computer placement and connections, Peripheral devices, Cables and network connections

14. a, b, and c. (Keylogging, data sniffing, and network logs.)

15. Commingling evidence means mixing personal and company data together, which can create legal complications.

16. Two common forensic hashing algorithms: MD5, SHA-1

17. False

18. True

19. You should take a live-acquisition field kit (for capturing volatile memory and running system data).

20. False

Creating & analyzing a forensic image

```
kali@kali: /images
File  Actions  Edit  View  Help
┌──(kali㉿kali)-[/]
└─$ ls
bin   etc     initrd.img      lib32       media  proc  sbin  sys  var
boot  home    initrd.img.old  lib64       mnt    root  srv   tmp  vmlinuz
dev   images  lib             lost+found  opt    run   swap  usr  vmlinuz.old

┌──(kali㉿kali)-[/]
└─$ cd images

┌──(kali㉿kali)-[/images]
└─$ ls
virtual_drive.E01  virtual_drive.info

┌──(kali㉿kali)-[/images]
└─$ mmls virtual_drive.E01
GUID Partition Table (EFI)
Offset Sector: 0
Units are in 512-byte sectors

      Slot      Start        End          Length       Description
000:  Meta      0000000000   0000000000   0000000001   Safety Table
001:  -------   0000000000   0000002047   0000002048   Unallocated
002:  Meta      0000000001   0000000001   0000000001   GPT Header
003:  Meta      0000000002   0000000033   0000000032   Partition Table
004:  000       0000002048   0000008158   0000006111   Microsoft basic data
005:  -------   0000008159   0000008191   0000000033   Unallocated

┌──(kali㉿kali)-[/images]
└─$
```

```
kali@kali: /images
File  Actions  Edit  View  Help
Cannot determine file system type

┌──(kali㉿kali)-[/images]
└─$ fsstat -o 2048 virtual_drive.E01
FILE SYSTEM INFORMATION
--------------------------------------------

File System Type: FAT12

OEM Name: mkfs.fat
Volume ID: 0xbdf013d
Volume Label (Boot Sector): NO NAME
Volume Label (Root Directory):
File System Type Label: FAT12

Sectors before file system: 2048

File System Layout (in sectors)
Total Range: 0 - 6110
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 5
* FAT 1: 6 - 10
* Data Area: 11 - 6110
** Root Directory: 11 - 42
** Cluster Area: 43 - 6110

METADATA INFORMATION
--------------------------------------------

Range: 2 - 97606
```

kali@kali: /images

File Actions Edit View Help

```
┌──(kali㉿kali)-[/images]
└─$ fls -o 2048 virtual_drive.E01
r/r 5:  singleton_3.jpg
r/r * 6:        _ECTURE2.JPG
r/r * 8:        lecture3.jpg
r/r * 10:       lecture4.jpg
r/r * 12:       lecture5.jpg
d/d 14: lecture
v/v 97603:      $MBR
v/v 97604:      $FAT1
v/v 97605:      $FAT2
V/V 97606:      $OrphanFiles

┌──(kali㉿kali)-[/images]
└─$ fls -o 2048 -rdp virtual_drive.E01
r/r * 6:        _ECTURE2.JPG
r/r * 8:        lecture3.jpg
r/r * 10:       lecture4.jpg
r/r * 12:       lecture5.jpg
```

Create A New Case

1. **Case Name:** The name of this investigation. It can contain only letters, numbers, and symbols.

   3000

2. **Description:** An optional, one line description of this case.

3. **Investigator Names:** The optional names (with no spaces) of the investigators for this case.

a. Andrew	b.
c.	d.
e.	f.
g.	h.
i.	j.

New Case Cancel Help

Adding host: `Virtual_Drive1` **to case** `3000`

Host Directory (`/var/lib/autopsy/3000/Virtual_Drive1/`) created

Configuration file (`/var/lib/autopsy/3000/Virtual_Drive1/host.aut`) created

We must now import an image file for this host

Add Image

Case: 3000
Host: Virtual_Drive1

ADD A NEW IMAGE

1. **Location**
Enter the full path (starting with /) to the image file.
If the image is split (either raw or EnCase), then enter '*' for the extension.

/images/virtual_drive.E01

2. **Type**
Please select if this image file is for a disk or a single partition.

(•) Disk ( ) Partition

3. **Import Method**
To analyze the image file, it must be located in the evidence locker. It can be imported from its current location using a symbolic link, by copying it, or by moving it. Note that if a system failure occurs during the move, then the image could become corrupt.

(•) Symlink ( ) Copy ( ) Move

NEXT

CANCEL HELP

Image File Details

Local Name: images/virtual_drive.E01

File System Details

Analysis of the image file shows the following partitions:

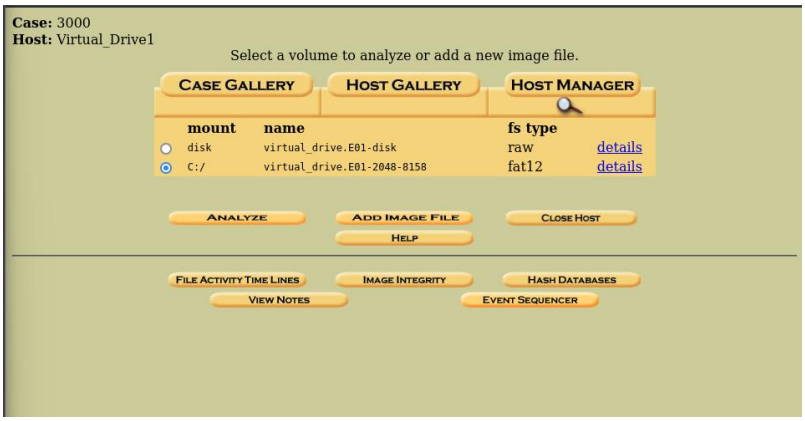
Partition 1 (Type: Microsoft basic data)
Sector Range: 2048 to 8158
Mount Point: C: File System Type: fat12

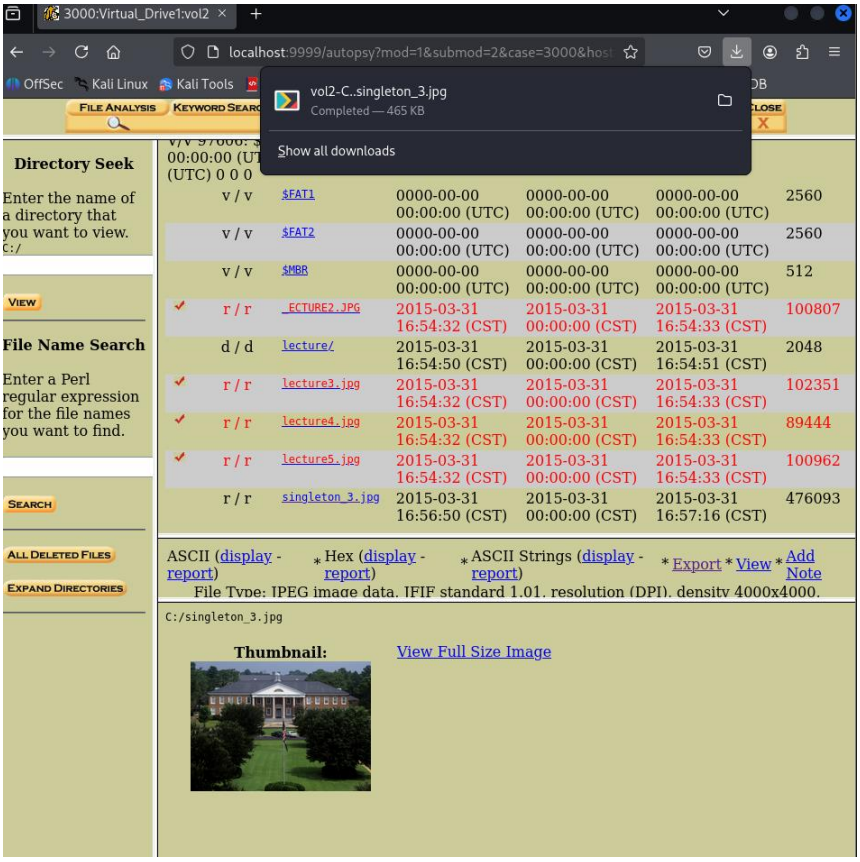
ADD CANCEL HELP

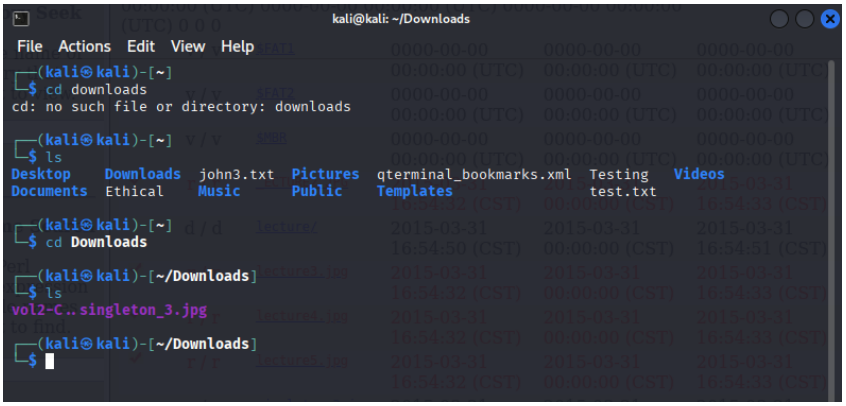
For your reference, the `mmls` output was the following:

```
GUID Partition Table (EFI)
Offset Sector: 0
Units are in 512-byte sectors

      Slot      Start        End          Length       Description
004:  000       0000002048   0000008158   0000006111   Microsoft basic data
```


Case: 3000
Host: Virtual_Drive1
Select a volume to analyze or add a new image file.
CASE GALLERY
HOST GALLERY
HOST MANAGER
mount name fs type
disk virtual_drive.E01-disk raw details
C:/ virtual_drive.E01-2048-8158 fat12 details
ANALYZE
ADD IMAGE FILE
CLOSE HOST
HELP
FILE ACTIVITY TIME LINES
IMAGE INTEGRITY
HASH DATABASES
VIEW NOTES
EVENT SEQUENCER


3000:Virtual_Drive1:vol2
localhost:9999/autopsy?mod=1&submod=2&case=3000&host
OffSec Kali Linux Kali Tools
vol2-C..singleton_3.jpg
Completed — 465 KB
Show all downloads
FILE ANALYSIS
Directory Seek
Enter the name of a directory that you want to view.
C:/
VIEW
File Name Search
Enter a Perl regular expression for the file names you want to find.
SEARCH
ALL DELETED FILES
EXPAND DIRECTORIES
$FAT1 0000-00-00 00:00:00 (UTC) 0000-00-00 00:00:00 (UTC) 0000-00-00 00:00:00 (UTC) 2560
$FAT2 0000-00-00 00:00:00 (UTC) 0000-00-00 00:00:00 (UTC) 0000-00-00 00:00:00 (UTC) 2560
$MBR 0000-00-00 00:00:00 (UTC) 0000-00-00 00:00:00 (UTC) 0000-00-00 00:00:00 (UTC) 512
r / r _ECTURE2.JPG 2015-03-31 16:54:32 (CST) 2015-03-31 00:00:00 (CST) 2015-03-31 16:54:33 (CST) 100807
d / d lecture/ 2015-03-31 16:54:50 (CST) 2015-03-31 00:00:00 (CST) 2015-03-31 16:54:51 (CST) 2048
r / r lecture3.jpg 2015-03-31 16:54:32 (CST) 2015-03-31 00:00:00 (CST) 2015-03-31 16:54:33 (CST) 102351
r / r lecture4.jpg 2015-03-31 16:54:32 (CST) 2015-03-31 00:00:00 (CST) 2015-03-31 16:54:33 (CST) 89444
r / r lecture5.jpg 2015-03-31 16:54:32 (CST) 2015-03-31 00:00:00 (CST) 2015-03-31 16:54:33 (CST) 100962
r / r singleton_3.jpg 2015-03-31 16:56:50 (CST) 2015-03-31 00:00:00 (CST) 2015-03-31 16:57:16 (CST) 476093
ASCII (display - report) * Hex (display - report) * ASCII Strings (display - report) * Export * View * Add Note
File Type: JPEG image data, JFIF standard 1.01, resolution (DPI), density 4000x4000,
C:/singleton_3.jpg
Thumbnail:
View Full Size Image


kali@kali: ~/Downloads
File Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ cd downloads
cd: no such file or directory: downloads
┌──(kali㉿kali)-[~]
└─$ ls
Desktop Downloads john3.txt Pictures qterminal_bookmarks.xml Testing Videos
Documents Ethical Music Public Templates test.txt
┌──(kali㉿kali)-[~]
└─$ cd Downloads
┌──(kali㉿kali)-[~/Downloads]
└─$ ls
vol2-C..singleton_3.jpg
┌──(kali㉿kali)-[~/Downloads]
└─$